Bessel's Correction $N \to N \to 1$

1. Centering introduces Linear Constraints

N data points:

$$x_1, x_2, \cdots, x_N$$

Centering:

$$x_i' = x_i - \bar{x}$$

Satisfies

$$\sum_{i=1}^{N} x_i' = 0 \Rightarrow \text{linear Constraint}$$

Centering $\to$ Orthogonal Projection

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix}, \quad \vec{1} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$$

$$\bar{x} = \frac{1}{N} \vec{x}^T \vec{1} = \frac{x_1 + x_2 + \cdots + x_N}{N} \to \text{scalar}$$

centering: $\vec{x}_c = \begin{bmatrix} x_1 - \bar{x} \\ x_2 - \bar{x} \\ \vdots \\ x_N - \bar{x} \end{bmatrix}$

$$= \vec{x} - \bar{x}\vec{1}$$

$$= \vec{x} - \frac{1}{N} \vec{x}^T \vec{1}\vec{1}$$

$$= \vec{x}\underbrace{(1 - \frac{1}{N}1^T 1)}_{\text{scalar}}$$

$$= (1 - \frac{1}{N}1^T 1)\vec{x}$$

$$\bar{X}_c = P\bar{X}$$

$$P = I - \frac{1}{N}11^T$$

$$\text{rank}(P) = N - 1$$

a. Satistics

samples: $\bar{X}_1, \bar{X}_2, \cdots, \bar{X}_N$

i.i.d

satisfies: $E[\bar{X}_i] = \mu$, $\text{Var}(\bar{X}_i) = \sigma^2$

sample mean: $E[\bar{X}] = \mu$

population mean: $\bar{X} = \frac{1}{N}\sum_{i=1}^{N} \bar{X}_i$

$$\sum_i (\bar{X}_i - \bar{\bar{X}})^2$$

$$= \sum_i \left[(\bar{X}_i - \mu) - (\bar{\bar{X}} - \mu)\right]^2$$

$$= \sum_i (\bar{X}_i - \mu)^2 + \sum_i (\bar{\bar{X}} - \mu)^2 - 2\sum_i (\bar{X}_i - \mu)(\bar{\bar{X}} - \mu)$$

$$= \sum_i (\bar{X}_i - \mu)^2 + N(\bar{\bar{X}} - \mu)^2 - 2N(\bar{\bar{X}} - \mu)^2$$

$$= \sum_i (\bar{X}_i - \mu)^2 - N(\bar{\bar{X}} - \mu)^2$$

$$\mathbb{E}\left[\sum_i (X_i - \bar{X})^2\right] = \mathbb{E}\left[\sum_i (X_i - \mu)^2\right] - n\mathbb{E}\left[(\bar{X} - \mu)^2\right]$$

$$= \sum_i \mathbb{E}\left[(X_i - \mu)^2\right] - n\,\mathbb{E}\left[(\bar{X} - \mathbb{E}[\bar{X}])^2\right]$$

$$= \sum_i \sigma^2 - n\,\mathrm{Var}(\bar{X}) \qquad (*)$$

$$\bar{X} = \frac{1}{n}\sum_i X_i$$

$$\mathrm{Var}(\bar{X}) = \frac{1}{n^2}\mathrm{Var}\left(\sum_i X_i\right) \quad \downarrow \text{independent (i.i.d)}$$

$$= \frac{1}{n^2}\sum_i \mathrm{Var}(X_i)$$

$$= \frac{1}{n^2}\, n\sigma^2$$

$$= \frac{\sigma^2}{n}$$

Sub into eqn $(*)$: $\mathbb{E}\left[\sum_i (X_i - \bar{X})^2\right] = (n-1)\sigma^2$

$$\Rightarrow \mathbb{E}\left[\frac{1}{n-1}\sum_i (X_i - \bar{X})\right] = \sigma^2$$

Def: $S^2 = \frac{1}{n-1}\sum_i (X_i - \bar{X})$ ← centering avg

satisfies: $\mathbb{E}[S^2] = \sigma^2$

Further Properties (mainly about $\bar{X}$ and μ)

(1) $\sum_{i=1}^{N}(X_i - \bar{X}) = 0$ $\Rightarrow$ no bias

(2) $\lim_{N\to\infty}(\bar{X} - \mu) = 0$

(3) $\varepsilon = \bar{X} - \mu$

$$\begin{cases} \mathbb{E}[\varepsilon] = 0 \\ \mathbb{E}[\varepsilon^2] = \frac{\sigma^2}{N} \end{cases}$$

$$\mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{1}{N}\sum_{i=1}^{N} X_i\right]$$

$$= \frac{1}{N}\sum_{i=1}^{N}\mathbb{E}[X_i]$$

$$= \frac{1}{N}\cdot N\mu$$

$$= \mu$$

$$\mathrm{Var}(\bar{X}) = \mathbb{E}\left[(\bar{X} - \mathbb{E}[\bar{X}])^2\right]$$

$$= \mathbb{E}\left[(\bar{X} - \mu)^2\right]$$

$$= \frac{\sigma^2}{N}$$